

§11.4 The Comparison Tests

Homework : 3,9,15,19,23,27,31,35,39,40,41,43,45,46

Who : $\sum_{n=1}^{\infty} a_n$, $a_n > 0$ for all n (或從某項以後恆正或恆負).

即對象為 essentially positive or negative series.

What : (I) 減法比較

$$(1) \ a_n \geq b_n \text{ and } \sum a_n \text{ converges} \Rightarrow \sum b_n \text{ converges.}$$

$$(2) \ a_n \geq b_n \text{ and } \sum b_n \text{ diverges} \Rightarrow \sum a_n \text{ diverges.}$$

(II) 除法比較

$$(1) \ \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0 \text{ (尾巴的速度相同).}$$

$$\Rightarrow \text{both } \sum a_n \text{ and } \sum b_n \text{ converge or diverge.}$$

$$(2) \ \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0 \text{ (} a_n \text{ 跑到 0 的速度比較快).}$$

$$(a) \ \sum b_n \text{ converges} \Rightarrow \sum a_n \text{ converges.}$$

$$(b) \ \sum a_n \text{ diverges} \Rightarrow \sum b_n \text{ diverges}$$

註：若 $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ (a_n 的尾巴速度比 b_n 快).

可得類似結論.

Example 1 :

	Series	Compared series	Conv./Div.
i.	$\sum \frac{1}{n^2 - n - 1}$	$\sum \frac{1}{n^2}$	Conv.
ii.	$\sum \frac{1}{\sqrt{n(n+1)(n+2)}}$	$\sum \frac{1}{n^{\frac{3}{2}}}$	Conv.

iii.	$\sum \frac{n^3}{2^n}$	$\sum \frac{1}{2^n}$	Conv.
iv.	$\sum \sin \frac{1}{n}$	$\sum \frac{1}{n}$	Div.
v.	$\sum \frac{n+5}{\sqrt[3]{n^7} + n^2}$	$\sum \frac{1}{n^{\frac{4}{3}}}$	Conv.
vi.	$\sum \frac{1}{n^{1+\frac{1}{n}}}$	$\sum \frac{1}{n}$	Div.
vii.	$\sum \frac{\ln n}{\sqrt{n}e^n}$	$\sum \frac{1}{e^n}$	Conv.

Example 2 : If $\sum a_n$ converges, where $a_n > 0$, is it true that $\sum \sin a_n$ is also convergent?

Solution :

Yes.

For (i) $a_n \rightarrow 0$ as $n \rightarrow \infty$.

$$(ii) \frac{\sin a_n}{a_n} \rightarrow 1 \text{ as } n \rightarrow \infty.$$

$$\Rightarrow \sum \sin a_n \text{ converges.}$$

Example 3 : If $\sum a_n$, $\sum b_n$ are both convergent series with positive terms, is it true that $\sum a_n b_n$ is also convergent?

Solution :

Yes.

For $a_n \rightarrow 0$ and $b_n \rightarrow 0$.

and $a_n b_n$ approaches 0 even faster than a_n (or b_n).

$$\Rightarrow \sum a_n b_n \text{ converges.}$$